

Revisit for AG,ANT and RT

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Chapter 1

Algebraic Geometry

Let us review concepts from Algebraic Geometry.

1. Let $A \in \text{Ring}$, then we have $\text{Sch}/_{\text{Spec}(A)} \simeq \text{Ring}_{A/}$. The correspondence is given by $(X \mapsto \text{Spec}(A)) \rightsquigarrow A \rightarrow \Gamma(X, \mathcal{O}_X)$.
2. If we consider $X \mapsto \mathbb{A}_A^1$, it corresponds to $A[t] \mapsto \Gamma(X, \mathcal{O}_X)$ which uniquely determined by the image of t . In this sense, we obtain that $(X \mapsto \mathbb{A}_A^1) \rightsquigarrow \Gamma(X, \mathcal{O}_X)$. Moreover, we have $\text{Hom}_S(X, \mathbb{A}_S^1) \simeq \text{Hom}_{\text{Spec}(\mathbb{Z})}(X, \mathbb{A}^1) \simeq \Gamma(X, \mathcal{O}_X)$.
3. Let A be a ring, we have $\text{Proj}/_{\text{Spec}(A)} \simeq \text{Mod}_A$.
4. Structure sheaf $\mathcal{O}_X : X^{\text{op}} \rightarrow \text{Ring}$.
5. Let $j_!$ denotes extension by zero and U is an open subscheme of X then we obtain that $\Gamma(U, \mathcal{F}) \cong \text{Hom}_{\mathcal{O}_X}(j_! \mathcal{O}_U, \mathcal{F})$.
6. Serre's criterion for Normality: A Noetherian ring A is normal if and only if it has properties R1 and S2. R1 means $A_{\mathfrak{p}}$ is regular(smooth) for any codimension one prime ideal $\mathfrak{p}$. S2 means $\text{depth}(A_{\mathfrak{p}}) = \min(2, \dim(A_{\mathfrak{p}}))$ for any prime ideal.
7. $\text{depth}(A_{\mathfrak{p}})$ means the maximal length of regular sequence $(a_1, a_2, \dots)$ in $\mathfrak{p}$. Regular means that a_i is not a zero divisor in $A/(a_1, \dots, a_i)$.
8. Sheaf of $\mathcal{O}_X$ -module:

$$\begin{array}{ccc}
 & & \text{Mod} \\
 & \nearrow \mathcal{F} & \downarrow P \\
 \text{Open}(X)^{\text{op}} & \xrightarrow{\mathcal{O}_X} & \text{Ring}
 \end{array}$$

9. $f : X \rightarrow Y$, then $f^* \mathcal{G} = f^{-1} \mathcal{G} \otimes_{f^{-1} \mathcal{O}_Y} \mathcal{O}_X$
10. Let M be A -module and $X = \text{Spec}(A)$. Then $\widetilde{M} := \underline{M} \otimes \mathcal{O}_X$ where $\underline{M}$ denotes the constant sheaf of M .
11. Ideal sheaf is the kernel of $i^\# : \mathcal{O}_X \rightarrow i_* \mathcal{O}_Y$ induced by closed immersion $i : Y \rightarrow X$.
12. We call a sheaf $\mathcal{F}$ is quasi-coherent if for any affine open subscheme U , there exist an $\mathcal{O}_X(U)$ -module M such that $\mathcal{F}|_U \simeq \widetilde{M}$. Hence, we call it coherent if X is Noetherian.
13. Let X be any affine scheme and $\mathcal{F}$ be an arbitrary quasi-coherent sheaf. Then we obtain that $H^1(X, \mathcal{F}) = 0$. It means that if we consider the exact sequence of ideal sheaf

$$0 \rightarrow \mathcal{I}_Y \rightarrow \mathcal{O}_X \rightarrow i_* \mathcal{O}_Y \rightarrow 0$$

The global section functor is exact

$$0 \rightarrow \Gamma(X, \mathcal{I}_Y) \rightarrow \Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X, i_* \mathcal{O}_Y) \rightarrow H^1(X, \mathcal{I}_Y) = 0.$$

Equivalently, we say any regular function on closed subscheme of X can be extended to X .

14. For any scheme X , S2 condition is equivalent to $j_*(\mathcal{O}_U) \simeq \mathcal{O}_X$ for any $\text{codim}(X \setminus U) \geq 2$
 15. For open subscheme U of X , there is an exact sequence

$$0 \rightarrow i_!\mathcal{O}_U \rightarrow \mathcal{O}_X \rightarrow j_*\mathcal{O}_Z \rightarrow 0$$

where $i_!\mathcal{O}_U$ is the push-forward with compact support. The inclusion is due to the extension by zero.

16. We also have exact sequence

$$0 \rightarrow \mathcal{H}_Z^0(\mathcal{O}_X) \rightarrow \mathcal{O}_X \rightarrow i_*\mathcal{O}_U \rightarrow \mathcal{H}_Z^1(\mathcal{O}_X) \rightarrow \dots$$

where $\mathcal{O}_X \rightarrow i_*\mathcal{O}_U$ is restriction. $\mathcal{H}_Z^0(\mathcal{O}_X)$ means the sheaf of supporting on Z . It also can be defined by the sheafication of presheaf functor $U \mapsto \Gamma_Z(X, \mathcal{O}_X) \cap U$ where Γ_Z means supporting on Z .

17. In the projective case, we have an similar exact sequence. Given that an projective scheme $X = \text{Proj}(S)$, then we interpret it as the open subscheme of CX excluding zero. The exact sequence follows

$$0 \rightarrow H_m^0(S) \rightarrow \Gamma(CX, \mathcal{O}_{CX}) = S \rightarrow \Gamma_*(\mathcal{O}_X) \rightarrow H_m^1(S) \rightarrow H^1(CX, \mathcal{O}_{CX}) = 0.$$

18. Let M be R -module, we define $\text{Ass}(M) = \{\text{Ann}_R(m)\}_{m \in M} \cap \text{Spec}(R)$. Then $\text{Supp}(M) = \bigcup_{\mathfrak{p} \in \text{Ass}(M)} V(\mathfrak{p})$. To see that we consider if $\mathfrak{p} = \text{Ann}_R(m)$ then $M_{\mathfrak{p}} \neq 0$.

19. Let $X = \text{Spec}(R)$ and $f \in R$, then $\text{Supp}(f) = V(\text{Ann}_R(f))$. Hence, if f is not zero divisor, then $\text{Supp}(f) = X$.

20. If we consider ${}_R R$, then $\text{Ass}(R) = \{\sqrt{\mathfrak{q}_1}, \dots, \sqrt{\mathfrak{q}_n}\}$ where $(0) = \mathfrak{q}_1 \cap \dots \cap \mathfrak{q}_n$ be the primary decomposition of zero ideal.

21. Suppose that I is radical and $Z = V(I)$. We have $\mathcal{H}_Z^0(\mathcal{O}_X) = \{x \in \mathcal{O}_X \mid \sqrt{\text{Ann}_R(x)} \supseteq I\}$. Hence, we can obtain that $\mathcal{H}_Z^0(\mathcal{O}_X) = \bigcap_{\mathfrak{p}_i \not\supseteq I} \mathfrak{q}_i$.

22. For a graded ring S with M is a graded S -module. Define $\widetilde{M}(D_+(f)) := \{m/f^n \mid \deg(m) = n\}$.

23. Let S be a graded ring, $S(n)$ denotes twisted module defined by $S(n)_d = S_{n+d}$. Define the twisted sheaf $\mathcal{O}_X(n) := \widetilde{S(n)}$ and twisted sheaf of module $\mathcal{F}(n) := \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.

24. Let S be the graded ring generated by S_1 as S_0 -algebra. Let $X = \text{Proj}(S)$. Then

- $\mathcal{O}_X(n)$ is an invertible sheaf on X ;
- $\widetilde{M}(n) \simeq \widetilde{M(n)}$;
- $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) \simeq \mathcal{O}_X(n+m)$

25. The section functor is given by $\Gamma_*(\mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F}(n))$. It is right adjoint functor of $\widetilde{(\cdot)}$. Hence if S is a graded ring generated by S_0 -algebra S_1 and $\mathcal{F}$ is quasi-coherent, then $\Gamma_*(\mathcal{F}) \simeq \mathcal{F}$.

26. Extension of Sections: Let X be a scheme and $\mathcal{L} \in \text{Pic}(X)$, $\mathcal{F} \in \text{Qcoh}(X)$. Suppose X is quasi-compact and has a finite open cover $\{U_i\}$ such that $\mathcal{L}|_{U_i}$ is free and $U_i \cap U_j$ is quasi compact. Given that $t \in \Gamma(X_f, \mathcal{F})$, then for some $n > 0$, the section $f^n t \in \Gamma(X_f, \mathcal{F} \otimes \mathcal{L}^{\otimes n})$ can be extended to a global section of $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$.

27. we call a morphism $f : X \rightarrow Y$ is projective if f can be factorized as form $X \rightarrow \mathbb{P}_Y^n \rightarrow Y$ where $X \rightarrow \mathbb{P}_Y^n$ is a closed immersion. In this case, we call X is a projective scheme over Y . If $Y = \text{Spec}(A)$, then exists a graded ring S finitely generated by $A = S_0$ -algebra S_1 .

28. we call a morphism is quasi projective if $X \rightarrow \mathbb{P}_Y^n$ is a locally closed immersion which means there exists closed subscheme Z such that $X \rightarrow Z$ is an open immersion.

- 29.** For any scheme Y , define twisting sheaf $\mathcal{O}(d)$ on $\mathbb{P}_Y^r$ by $g^*(\mathcal{O}(d))$ where $g : \mathbb{P}_Y^r \rightarrow \mathbb{P}_{\mathbb{Z}}^r$.
- 30.** X is a scheme over Y . We call invertible sheaf $\mathcal{L}$ is very ample relative to Y if there is a locally closed immersion $i : X \rightarrow \mathbb{P}_Y^r$ for some r such that $i^*(\mathcal{O}(1)) \simeq \mathcal{L}$.
- 31.** We call $\mathcal{F}$ is generated by global sections if there is a family of global sections $\{s_i\}$ such that $\mathcal{F}_x$ is generated by $\{s_i(x)\}$.
- 32.** Let A be a finitely generated k -algebra with graded structure. Let X be projective over $\text{Spec}(A)$ and $\mathcal{F}$ is an arbitrary coherent sheaf. Then $\Gamma(X, \mathcal{F})$ is a finitely generated A -module.
- 33.** Let Y is quasi-projective over $\text{Spec}(A)$. Equivalently, Y is a quasi-projective subscheme of $\mathbb{P}_A^n$. Then we have a very ample line bundle on Y .
- 34.** Flasque sheaf is a sheaf such that for any two $V \subseteq U$, we have

$$\rho_{U,V} : U \rightarrow V$$

is surjective. An example is consider $i : \text{pt} \rightarrow X$ and the sheaf $i_*(\mathcal{O}_{\text{pt}})$ is a Flasque sheaf of X .

- 35.** Every finite type affine k -scheme is quasi-projective.
- 36.** Let $X = \text{Spec } R$ with a sheaf of module $\tilde{M}$. Then we have

$$V(\text{Ann}_R(M)) = \text{supp}_X(\tilde{M}).$$

Let $R = k[x]$ and $R = k[x]/(x^2)$. Then the support of M is given by

$$Y = \text{Spec}(k[x]/(x^2))$$

But its topological support is Y_{red} .

- 37.** Let $B = A[x_1, \dots, x_n]/I$. The Kähler differential is given by

$$\Omega_{B/A} = \left(\bigoplus_{i=1}^n B dx_i \right) / \langle df, f \in I \rangle.$$

- 38.** Let $A \rightarrow B \rightarrow C$ be rings. There First exact sequence of Kähler differential is

$$\Omega_{B/A} \otimes_B C \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0$$

Actually, we have

$$\Omega_{B/A} = \Omega_{B/C} / \langle dg, g \in A \rangle$$

If we let $C = B/I$ then we obtain a second exact sequence

$$I/I^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \rightarrow \Omega_{C/A} \rightarrow 0.$$

Where I/I^2 is conormal module. The normal module is the dual of conormal module which define by

$$N_{C/A} = \text{Hom}_C(I/I^2, C).$$

The tangent bundle is given by

$$T_{C/A} = \text{Hom}_C(\Omega_{C/A}, C).$$

Let $X = \text{Spec}(A)$ and $Y = \text{Spec}(A/I)$ and smooth. We have the exact sequence

$$0 \rightarrow N_Y^{\vee} \xrightarrow{\delta} \Omega_X \otimes_{\mathcal{O}_X} \mathcal{O}_Y \rightarrow \Omega_Y \rightarrow 0.$$

where $N_{Y/X}^\vee$ denotes the sheaf induced by conormal module I/I^2 . Dualize we then have

$$0 \rightarrow T_Y \rightarrow T_X \otimes_{\mathcal{O}_X} \mathcal{O}_Y \rightarrow N_{Y/X} \rightarrow 0.$$

$$\begin{aligned} X &= \mathbb{A}_{\mathbb{C}}^2 = \operatorname{Spec} \mathbb{C}[x, y], \quad Y = V(y - x^2), \quad I = (y - x^2), \quad \mathcal{O}_Y \simeq \mathbb{C}[x]. \\ \Omega_X|_Y &= \mathcal{O}_Y dx \oplus \mathcal{O}_Y dy, \quad \Omega_Y \simeq (\mathcal{O}_Y dx \oplus \mathcal{O}_Y dy)/(dy - 2x dx), \quad T_Y = \mathcal{O}_Y (\partial_x + 2x \partial_y). \\ I/I^2 &\simeq d(y - x^2) = dy - 2x dx. \quad N_{Y/X} \simeq (\mathcal{O}_Y \partial_x \oplus \mathcal{O}_Y \partial_y)/\mathcal{O}_Y (\partial_x + 2x \partial_y). \end{aligned}$$

39. In local we also have an exact sequence

$$0 \rightarrow \mathfrak{m}_x/\mathfrak{m}_x^2 \rightarrow \mathcal{O}_{X,x} \rightarrow \mathcal{O}_{X,x}/\mathfrak{m}_x \rightarrow 0$$

where $\mathfrak{m}_x/\mathfrak{m}_x^2$ is fiber of conormal sheaf.

40. We define canonical sheaf of X by $\omega_X = \bigwedge^n \Omega_{X/k}$. If X is normal(integrally closed) let $\eta \in \omega_X$, we define canonical divisor by

$$K_X = \sum \operatorname{ord}_D(\eta) \cdot D.$$

Then $\mathcal{O}_X(K_X) = \omega_X$.

41. Let $H = V(x_0)$ be a hyperplane of $\mathbb{P}^n$. Then $\operatorname{Pic}(\mathbb{P}^n) = \langle H \rangle$.

42. Consider $\mathcal{O}_{\mathbb{P}^2}(-1)$ as tautology line bundle.

We actually have

$$\mathcal{O}(-1)(U_0) = \mathbb{C}[\frac{x_1}{x_0}, \frac{x_2}{x_0}] \cdot x_0^{-1}$$

One should notice that here x_0^{-1} just record the information of local frame. We can see that $x_1^{-1} = (x_1^{-1}x_0)x_0^{-1}$

43. Actually Consider $H^0(X, \mathcal{F})$ a k -linear space, the local frame for open subscheme U is a group of elements $\{x_1, \dots, x_n\} \subseteq \mathcal{F}(U)$ forms a $\mathcal{O}_X(U)$ -basis of $\mathcal{F}(U)$.

44. Let $Y = \operatorname{Spec} A$ and let $X = \mathbb{P}_A^n$. Then there is an exact sequence of sheaves.

$$0 \rightarrow \Omega_{X/Y} \rightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \rightarrow \mathcal{O}_X \rightarrow 0.$$

We can check by take $Y = \operatorname{Spec} \mathbb{C}$ and $X = \mathbb{P}_{\mathbb{C}}^2$. Consider the U_0 , then we obtain that

$$\begin{aligned} \Omega_{X/Y}(U_0) &\simeq \mathbb{C}[\frac{x_1}{x_0}, \frac{x_2}{x_0}]d(x_1/x_0) \oplus \mathbb{C}[\frac{x_1}{x_0}, \frac{x_2}{x_0}]d(x_2/x_0) \\ \mathcal{O}_X(-1)^{\oplus 3}(U_0) &\simeq \bigoplus_{i=1}^3 \mathbb{C}[\frac{x_1}{x_0}, \frac{x_2}{x_0}] \cdot x_0^{-1} \end{aligned}$$

The first homomorphism is given by

$$d(x_1/x_0) = d(x_1)/x_0 - x_1 d(x_0)/x_0^2 \mapsto (1, x_1/x_0, 0)$$

and similarly

$$d(x_2/x_0) \mapsto (1, 0, x_1/x_0).$$

The second homomorphism is

$$(f, g, h) \mapsto x_0^{-1}(f + g + h)$$

Then we can verify this sequence is exact over U_0 . The same as U_1, U_2 . Then we obtain that this sequence exact as a whole space.

There also have a exact sequence of tangent sheaf

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X(1)^{\oplus(n+1)} \rightarrow T_X \rightarrow 0$$

We call this type of sequence Euler sequence.

45. Euler sequence give a way to find canonical sheaf. We use the fact that

$$\det(\mathcal{O}_X(-1)^{\oplus(n+1)}) \simeq \det(\Omega_X) \otimes \det(\mathcal{O}_X)$$

Therefore we obtain that

$$\det(\Omega_X) = \omega_X = \mathcal{O}_X(-n-1).$$

46.

47. The Serre's Theorem for affiness. Let X be a Noetherian scheme. Then X is affine iff. $H^i(X, \mathcal{F}) = 0$ for all quasi-coherent sheaves and $i > 0$.

We first consider that $\mathbb{A}^2 \setminus \{0, 0\}$ with open cover $D(x), D(y)$. The Čech complex is given by

$$0 \rightarrow k[x, y]_x \oplus k[x, y]_y \xrightarrow{d} k[x, y]_{x, y} \rightarrow 0$$

where $d(f, g) = f - g$. Therefore, we find that

$$H^1(X, \mathcal{O}_X) = \frac{k[x, y]_{x, y}}{k[x, y]_x + k[x, y]_y}$$

We find that $[x^{-1}y^{-1}] \neq 0$.

48. The cohomolog of Projective space. Let we consider $X = \mathbb{P}_A^r$ and we know that

1. $H^i(X, \mathcal{O}_X(n)) = 0$ for $0 < i < r$;
2. $H^r(X, \mathcal{O}_X(-r-1)) \simeq A$;
3. $H^0(X, \mathcal{O}_X(n)) \times H^r(X, \mathcal{O}_X(-n-r-1)) \rightarrow H^r(X, \mathcal{O}_X(-r-1))$ is a perfect paring.

Let we check for $\mathbb{P}^2$ and sheaf $\mathcal{O}(1)$. We give cech complex

$$\begin{aligned} 0 \rightarrow \mathbb{C}[x_1/x_0, x_2/x_0] \cdot x_0 \oplus \mathbb{C}[x_0/x_1, x_2/x_1] \cdot x_1 \oplus \mathbb{C}[x_0/x_2, x_1/x_2] \cdot x_2 \\ \xrightarrow{d^0} \mathbb{C}[x_1/x_0, x_2/x_0, x_0/x_1] \cdot x_0 \oplus \mathbb{C}[x_1/x_0, x_2/x_0, x_0/x_2] \cdot x_0 \oplus \mathbb{C}[x_0/x_1, x_2/x_1, x_1/x_2] \cdot x_1 \\ \xrightarrow{d^1} \mathbb{C}[x_1/x_0, x_2/x_0, x_0/x_1, x_0/x_2] \cdot x_0 \\ \xrightarrow{d^2} 0 \end{aligned}$$

where we have

$$\begin{aligned} d^0(fx_0, gx_1, hx_2) &= ((x_1x_0^{-1}g - f) \cdot x_0, (x_2x_0^{-1}h - f) \cdot x_0, (x_2x_1^{-1}h - g) \cdot x_1) \\ d^1(fx_0, gx_0, hx_1) &= (f - g + x_1x_0^{-1}h) \cdot x_0 \end{aligned}$$

We consider

$$\begin{aligned} H^1(X, \mathcal{O}(1)) &= \frac{\langle (fx_0, fx_0, 0), (0, hx_1, hx_1) \rangle}{\langle (-fx_0, -fx_0, 0), (gx_1, 0, -gx_1), (0, hx_2, hx_2) \rangle} = 0 \\ H^2(X, \mathcal{O}(-3)) &= \frac{f \cdot x_0^{-3}}{\langle (f - g + x_1^{-3}x_0^3h) \cdot x_0^{-3} \rangle} \simeq \mathbb{C} \cdot \left[\frac{1}{x_0x_1x_2} \right] \end{aligned}$$

We also know that

$$H^0(X, \mathcal{O}_X(n)) \simeq \mathbb{C}[x_0, \dots, x_2]_n$$

$$H^2(X, \mathcal{O}_X(-n-r-1)) = \langle x_0^{a_0} x_1^{a_1} x_2^{a_2} \mid a_0 + a_1 + a_2 = -n \rangle_{\mathbb{C}}$$

The paring is given by multiplication.

49. Hartogs Extension Theorem. Consider U as a open subscheme of X let $Z = X \setminus U$. We let

$$\mathcal{H}_Z^0(\mathcal{F}) := \ker(\mathcal{F} \rightarrow j_* j^* \mathcal{F})$$

then we obtain a short exact sequence

$$0 \rightarrow \mathcal{H}_Z^0(\mathcal{O}_X) \rightarrow \mathcal{O}_X \rightarrow j_* \mathcal{O}_U$$

Acting on global section functor, we then have a long exact sequence

$$0 \rightarrow H_Z^0(X, \mathcal{O}_X) \rightarrow H^0(X, \mathcal{O}_X) \rightarrow H^0(X, j_* \mathcal{O}_U) \rightarrow H_Z^1(X, \mathcal{O}_X)$$

A global section of $\mathcal{O}_U$ can be extended to a global section of $\mathcal{O}_X$ is equivalent to say

$$H^0(X, \mathcal{O}_X) \rightarrow H^0(X, j_* \mathcal{O}_U)$$

is surjective. It require that

$$H_Z^1(X, \mathcal{O}_X) = 0.$$

When we require X is normal, then we obtain what we want.

50. Computation of Picard Group We have divisor sequence which is given by

$$0 \rightarrow \mathcal{O}_X^\times \rightarrow K_X^\times \rightarrow \text{Div}_X \rightarrow 0$$

It induces a long exact sequence

$$0 \rightarrow H^0(X, \mathcal{O}_X^\times) \rightarrow H^0(X, K_X^\times) \rightarrow H^0(X, \text{Div}_X) \rightarrow H^1(X, \mathcal{O}_X^\times) \rightarrow H^1(X, K_X^\times)$$

Since $K_X^\times$ is Flasque then it annihilate at H^1 . Then we obtain that

$$K_X^\times \rightarrow \text{Div}_X \rightarrow \text{Pic}(X) \rightarrow 0$$

Similarly, if we consider U is an open scheme of X , we can from this short exact sequence

$$0 \rightarrow \mathcal{O}_X^\times \rightarrow j_* \mathcal{O}_U^\times \rightarrow 0$$

51. Riemann-Roch Theorem

$$h^0(D) - h^0(K - D) = \deg(D) + 1 - g$$

We want to calculate $h^0(D)$.

When $X = \mathbb{P}^n$, we have $\deg(K - D) < 0$ then $h^0(K - D) = 0$.

Consider X be projective curve over $\mathbb{P}^2$, then we have

$$g = \frac{(d-1)(d-2)}{2}$$

and

$$K_X = \mathcal{O}_X(d - r - 1)$$

Since we have

$$\deg(\mathcal{O}_X(m)) = m \cdot \deg(X)$$

therefore we have $\deg(K_X) = 2g - 2 = d(d - 3)$.

If

$$\deg(K_X - D) < 0$$

then we have

$$h^0(D) = \deg(D) + 1 - g.$$

Example: let $D = P_1 + \cdots + P_k$ be k distinct divisor over projective curve X in $\mathbb{P}^2$ with degree 4. Prove that P_i on a same line if $h^0(X, D) = 2$, if not $h^0(X, D) = 1$.

Actually it means that there is a section s of $\mathcal{O}_X(1)$ such that $P_i \in \text{supp}(s)$. We have

$$K_X = \mathcal{O}_X(1)$$

Therefore we then must prove that

$$h^0(X, K_X - D) \geq 1$$

According to Riemann-Roch we then have

$$h^0(X, K_X - D) = h^0(X, D) - \deg(D) + 2$$

Chapter 2

Algebraic Number Theory

First let we consider some basic extensions:

Type A Quadratic Extensions $[L : K] = 2$;

Type B Cyclotomic Extensions $K(\zeta_n)/K$;

Type C Kummer Extensions $L = K(\sqrt[n]{a})$;

Type D Pure Cubic Extensions $L = K(\sqrt[3]{a})$;

Type E Cyclic Cubic Extensions $\text{Gal}(L/K) \cong C_3$;

Type F Biquadratic Extensions $L = K(\sqrt{a}, \sqrt{b})$;

Type G Quaternion Extensions $\text{Gal}(L/K) \cong Q_8$;

Type H Dihedral Extensions $\text{Gal}(L/K) \cong D_{2n}$;

Type I Complex Multiplication Fields $L = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$;

Type J Eisenstein Extension $L = \mathbb{Q}(\omega)$ where ω is a root of an Eisenstein polynomial.

Type K Artin-Schreier Extensions $L = K(y)$ with $y^p - y = a \in K$.

For Type C, the base field K must contains the roots of unity $\mu_n = \langle \zeta_n \rangle$. For Type K, $\text{char}(K) = p$.

Let us review concepts from Algebraic Number Theory.

Lattice over Dedekind domain

Lattice is the most important concept in Algebraic Number Theorem. Let we give some examples.

1. $\mathcal{O}_L$ as $\mathcal{O}_K$ -lattice equipped with $\text{Tr}_{L/K}$ as the bilinear form.
2. The Value Group $v(K^*)$ is a one-dimension lattice.
3. $L \otimes_K K_v$ is a K_v -vector space.
4. Residue Field $\kappa(\mathfrak{P})$ is a $\kappa(\mathfrak{p})$ -vector space.
5. The valuation of idele $\log(A_K^\times)$ produces lattice structure.

6. The Unit Lattice $\log : \mathcal{O}_K^\times \rightarrow \mathbb{R}^{r_1+r_2}$ and Minkowski Lattice $\mathcal{O}_K \rightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$.

There are several important concepts about lattice over Dedekind domain.

Definition 2.0.1 (Trace Ideal). The trace ideal $\mathfrak{T}_{L/K}$ is defined by scale ideal $\mathfrak{s}(\mathcal{O}_L) = \langle \text{Tr}_{L/K}(xy) \mid x, y \in \mathcal{O}_L \rangle_{\mathcal{O}_K}$.

Definition 2.0.2 (Discriminant of Field). The discriminant of field L is given by $\text{dics}(\langle \alpha_1, \dots, \alpha_n \rangle) = \det(\text{Tr}(\alpha_i \alpha_j))$ where $\{\alpha_i\}$ is an integral basis.

Definition 2.0.3 (Inverse Different and Different). The inverse different is given by the dual lattice $\mathcal{O}_L^\#$. And the different $\mathfrak{D}_{L/K}$ is $(\mathcal{O}_L^\#)^{-1}$.

Definition 2.0.4 (Different of an element). Let L/K be a finitely algebraic extension with a minimal polynomial $f(x)$. Then the different of an element α is $f'(\alpha) \cdot \mathcal{O}_L$.

Theorem 2.0.5 (Different Deduced by Different of Elements). We have an isomorphism

$$\mathfrak{D}_{L/K} \simeq (f'(\alpha))_{\alpha \in L}$$

When $\mathcal{O}_K$ is not PID, then $\mathcal{O}_L$ is not a free $\mathcal{O}_K$ -module. In this situation, we use Steinitz class to delineate it.

Theorem 2.0.6 (Steinitz Class). There exist a basis $\{e_1, \dots, e_n\}$ of L over K and fractional ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_n$ of $\mathcal{O}_K$ such that:

$$\mathcal{O}_L = \mathfrak{a}_1 e_1 \oplus \mathfrak{a}_2 e_2 \oplus \dots \oplus \mathfrak{a}_n e_n$$

Furthermore, for any non-zero $\mathcal{O}_L$ -ideal $\mathfrak{J}$, there exist fractional ideals $\mathfrak{b}_1, \dots, \mathfrak{b}_n$ of $\mathcal{O}_K$ such that:

$$\mathfrak{J} = \mathfrak{b}_1 e_1 \oplus \mathfrak{b}_2 e_2 \oplus \dots \oplus \mathfrak{b}_n e_n.$$

Definition 2.0.7 (Norm and Discriminant of an Ideal). The norm of an ideal $\mathfrak{J}$ is given by

$$N_{L/K}(\mathfrak{J}) = \frac{\mathfrak{b}_1 \cdots \mathfrak{b}_n}{\mathfrak{a}_1 \cdots \mathfrak{a}_n}.$$

The Discriminant of it is given by

$$\mathfrak{d}(\mathfrak{J}) = \det(\text{Tr}_{L/K}(e_i e_j)) \cdot (\mathfrak{b}_1 \cdots \mathfrak{b}_n)^2.$$

Actually, it is just the volume as an $\mathcal{O}_K$ -lattice.

Theorem 2.0.8 (Relation of Different and Discriminant). We have an isomorphism

$$\mathfrak{d}_{L/K} = N_{L/K}(\mathfrak{D}_{L/K}).$$

Theorem 2.0.9 (Discriminant of an order).

$$(\Delta_{L/K}(\theta)) = [\mathcal{O}_L : \mathcal{O}_K[\theta]]_{\mathcal{O}_K}^2 \cdot \Delta_{L/K}$$

Definition 2.0.10 (Conductor of an Order). Given an order $\mathcal{O}$, its conductor is given by

$$\mathfrak{f}(\mathcal{O}) = \{x \in \mathcal{O}_K \mid x \cdot \mathcal{O}_K \in \mathcal{O}\}.$$

Theorem 2.0.11 (Resultant and Field Norm for Non-Monic Minimal Polynomials). Let $K = \mathbb{Q}(\alpha)$ be a number field of degree n , and let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$$

be the minimal polynomial of α over $\mathbb{Q}$ with leading coefficient $a_n \neq 0$. Let $g(x) \in \mathbb{Q}[x]$ be a polynomial of degree m , and set $\beta = g(\alpha) \in K$.

Then the field norm of β over $\mathbb{Q}$ satisfies

$$N_{K/\mathbb{Q}}(\beta) = \frac{1}{a_n^m} \text{Res}(f, g),$$

where $\text{Res}(f, g)$ denotes the resultant of f and g .

Valuation

1. Consider algebraic number field $L/\mathbb{Q}$. Let $\mathfrak{p}$ lies over p . Then we have the ideal norm

$$N(\mathfrak{p}) = [\mathcal{O}_L : \mathfrak{p}] = \det(\mathfrak{p}/\mathcal{O}_L) = p^{f(\mathfrak{p})}.$$

where $f(\mathfrak{p})$ is inertia degree.

Local to Global Principle

Consider algebraic field extension L/K with a valuation v of K . Moreover we require L/K is finite and seperable. The Local-to-Global Principle is basically consider the K -embedding to $\bar{K}_v$ of L . Given by an embedding $\sigma : L \rightarrow \bar{K}_v$, we then have an induced valuation $w_\sigma := \bar{v} \circ \sigma$ where $\bar{v}$ is uniquely determined by v since K_v is Henselian. We let L_w denote the completion of L corresponding to the valuation w .

There are some facts about this process.

2. Every extension w of v is identified with $\bar{v} \circ \sigma$ for a embedding σ .
3. Two extension $\bar{v} \circ \sigma$ and $\bar{v} \circ \sigma'$ are equal if and only if σ and σ' are conjugate over K_v .
4. Suppose that $L = K(\alpha)$ with minimal polynomial $f(x)$. Then the extensions of valuation are 1-1 corresponds to the irreducible factors f_i in

$$f(x) = f_1(x) \cdots f_n(x)$$

of f over K_v .

5. we have a useful fact that

$$L \otimes K_v \simeq K[x]/f(x) \otimes_K K_v \simeq K_v[x]/f(x) \simeq \prod_{w|v} K_v[x]/f_w(x) \simeq \prod_{w|v} L_w.$$

We should notice that the decomposition $f(x) = \prod f_w(x)$ relies on seperality of L/K .

6. We also have the norm and trace formula

$$N_{L/K}(\alpha) = \prod_{w|v} N_{L_w/K_v}(\alpha) \text{ and } \text{Tr}_{L/K}(\alpha) = \sum_{w|v} \text{Tr}_{L_w/K_v}(\alpha).$$

Newton Polygon and Hensel's Lemma

Definition 2.0.12 (Newton Polygon). Let K be a field with a discrete valuation v , and let

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \in K[x]$$

be a polynomial with $a_0, a_n \neq 0$. The *Newton polygon* of f with respect to v is the lower convex hull of the set of points

$$\{(i, v(a_i)) : 0 \leq i \leq n, a_i \neq 0\} \subset \mathbb{R}^2.$$

Theorem 2.0.13 (Newton Polygon Decomposition). Let (K, v) be a Henselian valued field and $f(x) \in K[x]$ a monic polynomial. Suppose the Newton polygon of f consists of segments with slopes $-\lambda_1 < -\lambda_2 < \cdots < -\lambda_r$ and horizontal lengths $\ell_1, \ell_2, \dots, \ell_r$. Then f factors uniquely in $K[x]$ as

$$f(x) = f_1(x) \cdot f_2(x) \cdots f_r(x),$$

where each $f_i(x) \in K[x]$ is monic of degree ℓ_i , and every root of f_i in $\bar{K}$ has valuation λ_i .

Lemma 2.0.14 (Hensel's Lemma). If a primitive polynomial $f(x) \in \mathcal{O}[x]$ admits modulo $\mathfrak{p}$ a factorization

$$f(x) \equiv \bar{g}(x)\bar{h}(x) \pmod{\mathfrak{p}}$$

into relatively prime polynomials $\bar{g}, \bar{h} \in \kappa[x]$, then $f(x)$ admits a factorization

$$f(x) = g(x)h(x)$$

into polynomials $g, h \in \mathcal{O}[x]$ such that $\deg(g) = \deg(\bar{g})$ and

$$g(x) \equiv \bar{g}(x) \pmod{\mathfrak{p}} \quad \text{and} \quad h(x) \equiv \bar{h}(x) \pmod{\mathfrak{p}}.$$

There is a very useful corollary about Hensel's lemma.

Corollary 2.0.15 (Lift of Simple root). Suppose that K is a henselian field with a valuation v . Let $\kappa(\mathfrak{p})$ denote the residue field of $\mathcal{O}_v$. Let $f(x) \in \mathcal{O}_v[x]$ and $\bar{f} = f \pmod{\mathfrak{p}}$. If there exists $\bar{a} \in \kappa(\mathfrak{p})$ such that

$$\bar{f}(\bar{a}) = 0 \text{ and } \bar{f}'(\bar{a}) \neq 0.$$

Then there exists a unique lift $a \in \mathcal{O}_v$ such that $\bar{a} = a \pmod{\mathfrak{p}}$ and $f(a) = 0$.

Theorem 2.0.16 (Dumas's Criterion). Let K be a field equipped with a discrete valuation v . Let

$$f(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in K[x]$$

be a polynomial of degree n with $a_na_0 \neq 0$. If the Newton polygon of $f(x)$ consists of a single line segment connecting the points $(0, v(a_0))$ and $(n, v(a_n))$, and if the horizontal length and vertical length of this segment are coprime, i.e.,

$$\gcd(n, |v(a_n) - v(a_0)|) = 1,$$

then $f(x)$ is irreducible over K .

Ramification

Definition 2.0.17 (Tamely Ramification). We call $\mathfrak{P}$ is tamely ramified over $\mathfrak{p}$ if $\kappa(\mathfrak{P})/\kappa(\mathfrak{p})$ is separable and $\gcd(e(\mathfrak{P}/\mathfrak{p}), \text{char}(\kappa(\mathfrak{p}))) = 1$.

We give a equivalent description when K is Henselian.

Theorem 2.0.18 (Tamely Ramification). A finite extension $L|K$ is tamely ramified if and only if the extension $L|T$ is generated by radicals

$$L = T(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r})$$

such that $(m_i, p) = 1$. In this case the fundamental identity always holds:

$$[L : K] = ef.$$

Proposition 2.0.19 (Different and Tame Ramification). Let $L|K$ be a finite separable extension of Henselian fields. Then

$$v_L(\mathfrak{d}_{L|K}) \geq e - 1,$$

with equality if and only if $\mathfrak{P}/\mathfrak{p}$ is tamely ramified:

$$\mathfrak{P}/\mathfrak{p} \text{ is tamely ramified} \iff v_L(\mathfrak{d}_{L|K}) = e - 1.$$

Equivalently, in terms of ideals,

$$\mathfrak{d}_{L|K} = \mathfrak{P}^{e-1} \iff \mathfrak{P}/\mathfrak{p} \text{ is tamely ramified.}$$

In the wildly ramified case, $v_L(\mathfrak{d}_{L|K}) \geq e$.

Proposition 2.0.20 (Trace and Tame Ramification). The extension is tamely ramified if and only if the trace ideal $\mathfrak{T}_{L/F} = \mathcal{O}_K$.

Theorem 2.0.21 (Dedekind-Kummer Theorem). Assume $L = K(\alpha)$ and $\alpha \in \mathcal{O}_L$, let $f \in \mathcal{O}_K[x]$ be the minimal polynomial of α . Assume that $\text{char}(\kappa(\mathfrak{p})) \nmid N(\mathcal{O}_K[\alpha])$. Suppose $g_1, \dots, g_r \in \mathcal{O}_K[x]$ are monic polynomials for which

$$\bar{f} = \bar{g}_1^{e_1} \cdots \bar{g}_r^{e_r}$$

is a complete factorization of $\bar{f} \in (\mathcal{O}_K/\mathfrak{p})[x]$, where $\bar{\cdot}$ denotes reduction modulo $\mathfrak{p}$, and let $\mathfrak{q}_i := (\mathfrak{p}, g_i(\alpha))$ be the $\mathcal{O}_L$ -ideal generated by $\mathfrak{p}$ and $g_i(\alpha)$. Then

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_r^{e_r},$$

is the prime factorization of $\mathfrak{p}\mathcal{O}_L$ in $\mathcal{O}_L$ and the residue degree of $\mathfrak{q}_i$ is $f_i := \deg g_i$.

The condition $\text{char}(\kappa(\mathfrak{p})) \nmid N(\mathcal{O}_K[\alpha])$ holds iff. any prime ideal $\mathfrak{P}$ lying over $\mathfrak{p}$ not contains conductor $\mathfrak{f}$. Therefore we have

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \mathcal{O}_{K,\mathfrak{p}} \simeq \mathcal{O}_K[\alpha] \otimes_{\mathcal{O}_K} \mathcal{O}_{K,\mathfrak{p}}$$

Tensor product each side by $\hat{\mathcal{O}}_{K,\mathfrak{p}}$, then we obtain that

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \hat{\mathcal{O}}_{K,\mathfrak{p}} \simeq \mathcal{O}_K[\alpha] \otimes_{\mathcal{O}_K} \hat{\mathcal{O}}_{K,\mathfrak{p}} \simeq \hat{\mathcal{O}}_{K,\mathfrak{p}}[x]/f(x).$$

Since $\hat{\mathcal{O}}_{K,\mathfrak{p}}$ shares the same residue field with $\mathcal{O}_K$ over $\mathfrak{p}$, the decomposition of $\bar{f}$ then can be lift to the local ring $\hat{\mathcal{O}}_{K,\mathfrak{p}}$. Using CRT on the right hand side, we then have

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \hat{\mathcal{O}}_{K,\mathfrak{p}} \simeq \prod_{i=1}^r \hat{\mathcal{O}}_{K,\mathfrak{p}}[x]/(g_i^{e_i})$$

There for we just need to consider the decomposition of $\mathfrak{p}$ on each $\hat{\mathcal{O}}_{K,\mathfrak{p}}[x]/(g_i^{e_i})$.
maximal unramif Kummer Theory

Ideal Class Group

We considier algebraic number field $L/\mathbb{Q}$.

7. The Minkowski embedding just given by canonical embedding $L \rightarrow L \otimes \mathbb{R}$. Given a primitive element and minimal polynomial, it can be explicitly described by

$$\mathbb{R}[x]/f(x) \simeq \bigoplus_{i=1}^{r_1} \mathbb{R}[x]/(x - a_i) \oplus \bigoplus_{j=1}^{r_2} \mathbb{R}[x]/(x^2 + b_j x + c_j) = \bigoplus_{i=1}^{r_1} \mathbb{R} \oplus \bigoplus_{j=1}^{r_2} \mathbb{C}$$

Therefore, the image of the ring of integers $\mathcal{O}_L$ under this embedding forms a $\mathbb{Z}$ -lattice of rank $r_1 + 2r_2$ inside this space.

8. Dirichlet Unit Theorem: $K^* \simeq \pi(K) \times \mathbb{Z}^{r_1+r_2-1}$.

9. The Minkowski constant of L is given by

$$M(L) = \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} |d(K)|^{1/2}$$

A quick method to find the roots of unity for $\mathcal{O}_K$. Consider $\mathfrak{p}$ is a prime ideal with residue field $\mathbb{F}_q$. Then the canonical isomorphism impose that

$$|W_K| \mid q - 1.$$

Therefore, we can find some prime ideals to minimize our search.

Hilbert's Ramification Tower

Definition 2.0.22 (Decomposition Group and Decomposition Field). If $\mathfrak{P}$ is a prime ideal of $\mathcal{O}$, then the subgroup

$$G_{\mathfrak{P}} = \{\sigma \mid \sigma(\mathfrak{P}) = \mathfrak{P}\}$$

is called the decomposition group of $\mathfrak{P}$ over K . The fixed field $Z_{\mathfrak{P}}$ is called the decomposition field.

Definition 2.0.23 (Inertia Group and Inertia Field). The Inertia group of $\mathfrak{P}$ is given by the kernel of homomorphism

$$G_{\mathfrak{P}} \rightarrow G(\kappa(\mathfrak{P})/\kappa(\mathfrak{p})).$$

The fixed field $I_{\mathfrak{P}}$ is called the inertia field.

We have a tower of field extension by

$$K \xrightarrow{\text{split}} Z_{\mathfrak{P}} \xrightarrow{\text{inertia extension}} T_{\mathfrak{P}} \xrightarrow{\text{ramification}} L$$

If L/K is unramified, it means that $I_{\mathfrak{P}}$ is trivial and thus we have an isomorphism

$$G_{\mathfrak{P}} \simeq G(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$$

Definition 2.0.24 (Frobenius Element). Let L/K be a Galois extension of number fields, and let $\mathfrak{P}$ be a prime of L lying over $\mathfrak{p}$ of K . Assume $\mathfrak{P}$ is unramified over $\mathfrak{p}$. The Frobenius element $\text{Frob}_{\mathfrak{P}}$ is the unique element in the Galois group $\text{Gal}(L/K)$ such that for all $\alpha \in \mathcal{O}_L$:

$$\text{Frob}_{\mathfrak{P}}(\alpha) \equiv \alpha^{N_{K/\mathbb{Q}}(\mathfrak{p})} \pmod{\mathfrak{P}}$$

The Frobenius map of residue field over $\mathfrak{P}$ is defined by $x \mapsto x^{N_{K/\mathbb{Q}}(\mathfrak{p})}$ since we want to fix the base field K .

Classification of extension over a local field

Suppose we consider the algebraic extension $L/\mathbb{Q}_p$ with degree n . The degree admits a decomposition

$$n = f \cdot e_0 \cdot p^k.$$

The inertia degree f corresponds to unramified extensions $L^{\text{umf}}/\mathbb{Q}_p$. And we have

$$L^{\text{umf}} = \mathbb{Q}_p(\zeta_{p^f-1}).$$

It is uniquely determined.

The tamely ramified index e_0 corresponds to tamely ramified extensions L^1/L^{umf} . It has the form

$$L^1 \simeq L^{\text{umf}}(\sqrt[e_0]{\pi})$$

where π is the uniformizer of L^{umf} . If $L^{\text{umf}} = \mathbb{Q}_p$, the uniformizer is exactly p .

The residue index p^k corresponds to wildly ramified extensions L/L^1 .

Actually, we can see more explicitly from the decomposition

$$\mathbb{Q}_p^\times \simeq \mathbb{Z} \times \mathbb{Z}_p^\times \simeq \mathbb{Z} \times (1 + p\mathbb{Z}_p) \times \mu_{p-1}$$

This decomposition is considered by canonical homomorphism

$$\mathbb{Z}_p \rightarrow \mathbb{F}_p$$

which induces

$$\mathbb{Z}_p^\times / (1 + p\mathbb{Z}_p) \simeq \mathbb{F}_p^\times.$$

Class Field Theory

Definition 2.0.25 (Local Artin Map). Let K be a local field, then there is a canonical homomorphism

$$\theta_K : K^\times \rightarrow \text{Gal}(K^{\text{ab}}/K).$$

where K^{ab} denotes the maximal abelian extension of K .

The general construction of local artin map is quite complicated. Mainly, it has two different way. The one is Lubin-Tate formal group and the other is Tate cohomology. We focus on cohomological way later.

Now let we consider a finite abelian extension L/K over local field K . The Artin map induces a canonical epimorphism

$$\theta_{L/K} : K^\times \rightarrow \text{Gal}(L/K).$$

with kernel $N_{L/K}(K^\times)$. Therefore, we have an isomorphism

$$K^\times / N_{L/K}(K^\times) \simeq \text{Gal}(L/K).$$

Definition 2.0.26 (Idele and Idele class group). The Idele group of a field V is defined by

$$\mathbb{I}_L = \prod'_v L_v^\times = \{(\alpha_v)_v \mid \text{All but finitely many } \alpha_v \in \mathcal{O}_v^\times\}.$$

Idele class group C_L is defined by $\mathbb{I}_L / L^\times$.

If we consider the field extension L/K , then we have

$$\mathbb{I}_L = \prod'_v \left(\prod_{w|v} L_w^\times \right)$$

Definition 2.0.27 (Global Artin Map). The global artin map of a global field K is given by

$$\begin{aligned} \theta_K : \mathbb{I}_K &\rightarrow \text{Gal}(K^{\text{ab}}/K) \\ (\alpha_v)_v &\mapsto \prod_v \theta_{K_v}(\alpha_v) \end{aligned}$$

Since $\mathbb{I}_K$ is defined by restricted direct product, our definition is well-defined.

Definition 2.0.28 (Relative Artin Map). Consider a finite abelian extension L/K of a global field K . The Relative Artin map is given by

$$\begin{aligned} \theta_{L/K} : \mathbb{I}_K &\rightarrow \text{Gal}(L/K) \\ (\alpha_v)_v &\mapsto \prod_v \theta_{L_v/K_v}(\alpha_v). \end{aligned}$$

Theorem 2.0.29 (Artin Reciprocity Law). The relative Artin map of L/K is surjective and with kernel

$$\ker(\theta_{L/K}) = K^\times \cdot N_{L/K}(\mathbb{I}_L).$$

It induces an isomorphism

$$\mathbb{I}_K / (K^\times \cdot N_{L/K}(\mathbb{I}_L)) \simeq \text{Gal}(L/K).$$

We can also write

$$C_K / N_{L/K}(C_L) \simeq \text{Gal}(L/K).$$

Now we consider the "the functor of point" perspective in Class Field Theory. Mainly, we interpret an element $a \in K$ as the representation of absolute galois group G_K .

To show that, we consider Kummer Theory. Let K be a field contains μ_n . Let a be an arbitrary nonzero element of K . We consider the splitting field of $x^n - a$. In algebraically closed field $\bar{K}$, we must have the decomposition

$$(x - \sqrt[n]{a})(x - \zeta_n \cdot \sqrt[n]{a}) \cdots (x - \zeta_n^{n-1} \cdot \sqrt[n]{a})$$

Let σ be an arbitrary element of G_K , then we consider

$$\frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} = \zeta^k$$

for some k . Actually, we use a type of polynomials $x^n - y$ induces the character

$$K^\times \rightarrow \text{Hom}_{\text{grp}}(G_K, \mu_n).$$

Actually, if we consider galois cohomology, then G_K act trivially on μ_n since $\mu_n \subseteq K$. In this sense we have

$$H^1(G_K, \mu_n) \simeq \text{Hom}_{\text{cont}}(G_K, \mu_n)$$

Then kummer exact sequence

$$1 \rightarrow \mu_n(K) \rightarrow K^\times \xrightarrow{(\cdot)^n} K^\times \rightarrow 1$$

induces a long exact sequence

$$1 \rightarrow \mu_n(K) \rightarrow K^\times \xrightarrow{(\cdot)^n} K^\times \rightarrow H^1(G_K, \mu_n(K)) \rightarrow H^1(G_K, K^\times) \rightarrow \cdots$$

Hilbert's 90 implies that $H^1(G_K, K^\times) = 0$ then we have the following corollary.

Corollary 2.0.30 (Kummer Isomorphism). Let K be an arbitrary field contains μ_n . Then we have an isomorphism

$$K^\times / K^{\times n} \simeq H^1(G_K, \mu_n) \simeq \text{Hom}_{\text{cont}}(G_K, \mu_n).$$

We have bijection

$$\{\text{Finite abelian extension } L/K\} \leftrightarrow \{\text{Finite subgroups of } G_K^\vee\}$$

Another G_K -module we usually consider in Class Field Theory is $\bar{K}^\times$. Here, we must notice that K is a local field. So what the element of K stand for in this sense? Let we consider K acts on the Pontryagin dual group $G_K^\vee$, which defined by

$$\begin{aligned} a : G_K^\vee &\rightarrow \mathbb{Q}/\mathbb{Z} \\ \chi &\mapsto \chi(\theta_K(a)). \end{aligned}$$

where θ_K means artin map.

Actually, it is the Tate-Nakayami duality. We first intepret a as an element of $H^0(G_K, \bar{K}^\times)$ and χ as an element

$$H^1(G_K, \mathbb{Q}/\mathbb{Z}) \simeq G_K^\vee.$$

The exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

We notice that $H^i(G_K, \mathbb{Q})$ since $H^i(G, \mathbb{Q}) = 0$ for $i > 0$ and any finite group G . Then know that

$$H^1(G_K, \mathbb{Q}/\mathbb{Z}) \simeq H^2(G_K, \mathbb{Z}).$$

We consider the cup product

$$H^2(G_K, \mathbb{Z}) \cup H^0(G_K, \bar{K}^\times) \rightarrow H^2(G_K, \mathbb{Z} \otimes_{\mathbb{Z}} \bar{K}^\times)$$

Therefore we have

$$\delta(\chi) \cup a \in H^2(G_K, \bar{K}^\times)$$

The Hasse invariant map tells

$$\text{inv} : H^2(G_K, \bar{K}^\times) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}.$$

Now we turns to the global situation. Given an element $a \in \mathbb{I}_K$ and $\chi \in G_K^\vee$ we define pairing

$$\langle \chi, (a_v)_v \rangle := \sum_v \text{inv}_v(\delta(\chi_v) \cup a_v) \in \mathbb{Q}/\mathbb{Z}$$

We also call the second cohomology group as Brauer group $\text{Br}(K) = H^2(G_K, \bar{K}^\times)$. An important exact sequence to give the well-definedness of this pairing is following theorem.

Theorem 2.0.31 (Hasse exact sequence). Let we consider global field K . Then we have a exact sequence of Brauer group

$$0 \rightarrow \text{Br}(K) \rightarrow \bigoplus_v \text{Br}(K_v) \xrightarrow{\sum_v \text{inv}} \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

The exact sequence implies that if $(a_v)_v$ is principle i.e. $(a_v)_v \in K^\times$, we then have

$$\langle \chi, (a_v)_v \rangle = 0.$$

Then we obtain that the pairing of global field

$$C_K \times H^1(G_K, \mathbb{Q}/\mathbb{Z}) \rightarrow \mathbb{Q}/\mathbb{Z}.$$

Now let we return to Kummer theorem again. Let K be a field containing μ_n . We can interpret μ_n as torsion submodule of $\bar{K}^\times$.

In this case, the Hilbert symbol over K_v is defined by

$$(\chi_a, b)_v \mapsto \frac{\theta_b(\sqrt[n]{a})}{\sqrt[n]{a}}$$

where $\chi_a \in K_v^\times / (K_v^\times)^n$ is Kummer character given by Kummer exact sequence and b is an element of K_v . θ_b is $\theta(b)$ the image of local artin map.

The global case of Hilbert reciprocity law is

$$\prod_v (a, b)_v = 1$$

Now we give the definition of local artin map over $\mathbb{Q}_p$. Consider the finite abelian extension $K/\mathbb{Q}_p$. Let $x = up^m \in \mathbb{Q}_p^\times$. The artin map is given by

$$\theta_{\mathbb{Q}_p}(p) = \text{Frob}_p.$$

and

$$\theta_{\mathbb{Q}_p}(u)(\zeta_{p^r}) = \zeta_{p^r}^{u \bmod p^r}$$

The calculation of quadratic Hilbert symbol. Let p be odd prime number. Consider $\mathbb{Q}_p$ and $a = p^\alpha u, b = p^\beta v$. Then the Hilbert symbol is given by

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right)^\beta \left(\frac{\bar{v}}{p}\right)^\alpha$$

where $\bar{u}, \bar{v} \in \mathbb{F}_p$. The legendre symbol is defined by

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p$$

We also have

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

The reciprocity law

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Also we should know

$$(a, b)_\infty = -1 \text{ if } a, b < 0$$

The Hilbert symbol has good properties:

$$\text{bilinear } (aa', b) = (a, b)(a', b)$$

$$\text{skew-symmetric } (a, b)(b, a) = 1$$

$$\text{non-degenerated } (a, b) = 1 \text{ for all } b \text{ iff. } a \in K^{\times n}$$

Problems

1. Let D be a square-free integer, $D \neq 0, 1$. For a prime number l , prove that if $l \mid D$, then l is ramified in $\mathbb{Q}(\sqrt{D})$.

Since D be a square-free integer and $l \mid D$, then we must have $v(D) = 1$. Then we have $v(\sqrt{D}) = 1/2$ implies it is ramified.

2. Analysis Cyclotomic extension $\mathbb{Q}(\zeta_n)$.

The cyclotomic polynomial are given by

$$\Phi(x) = \sum_{k=1, (n,k)=1}^n (x - \zeta_n^k).$$

Therefore $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$. The Euler function $\varphi(p^k) = p^k - p^{k-1}$. The galois group of this extension is given by $(\mathbb{Z}/n\mathbb{Z})^\times$. We need to notice that $|(\mathbb{Z}/p^k\mathbb{Z})^\times| = \varphi(p^k)$.

3. Every quadratic extension over $\mathbb{Q}$ must have the form $\mathbb{Q}(\sqrt{D})$. It is Galois and its roots of unity only have three types.

Clearly, Consider $\mathbb{Q}(\alpha)$ where α with minimal polynomial $f(x)$. According to the Vieta's formulas, we must have $\beta = \alpha^{-1}c/a$. It is the split field of $f(x)$ and hence it is Galois.

The Galois group must have order two, it should be $\mathbb{Z}/2\mathbb{Z}$.

Let D be the square-free part of the square of determinant $\sqrt{\Delta}$ of $f(x)$, then we can check that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{D})$.

Suppose that $\mathbb{Q}(\zeta_{p^m}) \subseteq \mathbb{Q}(\alpha)$. Since $[\mathbb{Q}(\zeta_{p^m}) : \mathbb{Q}] = \varphi(p^m) = p^{m-1}(p-1)$. Then inequality

$$p^{m-1}(p-1) \leq 2$$

tell us it only can be the case:

1. $p = 3, m = 1$.
2. $p = 2, m = 1$.
3. $p = 2, m = 2$.

It corresponds to the roots of unity W_k

1. $\langle -1, \omega, \omega^2 \rangle$
2. $\langle -1 \rangle$
3. $\langle -1, i \rangle$

Clearly $D > 0$ then we have $W_k = \langle -1 \rangle$. If $D = -3$, then we find that $\mathbb{Q}(\frac{\sqrt{-3}+1}{2}) = \mathbb{Q}(\zeta_3)$. If $D = -1$, it must be $\mathbb{Q}(i)$. And others situation must have root of unity $\langle -1 \rangle$.

4. The roots of unity over $\mathbb{F}_{p^t}$.

Since every element $x \in \mathbb{F}_{p^t}$ we have $x^{p^t-1} = 1$. Then $x^n = 1$ and $(n, p) = 1$ iff. $n | p^t - 1$. If $p | n$, then we find that $x^{kp^s} - 1 = (x^k - 1)^{p^s} = 0$.

5. Kronecker-Weber Theorem state that every Abel number field is a subfield of a cyclotomic number field. Find this field for quadratic extension $\mathbb{Q}(\sqrt{D})$.

Let we consider the determinant Δ of $f(x)$ which is the minimal polynomial of $\sqrt{D}$. Then consider $\mathbb{Q}(\zeta_{|\Delta|})$.

6. Classical Group in Galois Theory.

There are several classical group we need to know.

1. Symmetric group S_n
2. Alternating Group A_n
3. Cyclic Groups $\mathbb{Z}/n\mathbb{Z}$
4. Dihedral Groups $D_n = \mathbb{Z}/n\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$.
5. Klein Four Group $V_4 = (\mathbb{Z}/2\mathbb{Z})^2$

6. Frobenius Groups $\text{AGL}(1, p) = \mathbb{Z}/p\mathbb{Z} \rtimes (\mathbb{Z}/p\mathbb{Z})^\times$.

7. Discriminant of $x^n + ax + b$.

We use the fact that $\Delta(f) = (-1)^{n(n-1)/2} \text{Res}(f, f')$. And

$$\text{Res}(f, f') = (-1)^{n-1} (n-1)^{n-1} a^n + n^n b^{n-1}.$$

8. Consider $\mathbb{Q}(\alpha)$ and with $f(x) = x^3 - x + 1$. Calculate the norm of $\alpha + 2$.

We compute the discriminant of f by

$$\Delta(f) = (-1) \text{Res}(f, f') = -23.$$

Let $g(x) = x + 2$, then we have

$$N_{L/K}(\alpha + 2) = \text{Res}(f, g).$$

Also we can find the splitting field of $f(x)$ is $\mathbb{Q}(\alpha, \sqrt{23})$. Then $\sqrt{23} \cdot \mathcal{O}_K$ lying over $23 \cdot \mathbb{Q}$. We next compute the

$$N_{L/K}(\alpha) = \text{Res}(f, x) = -1.$$

9. Verify the integral closure of $\mathbb{Z}[\alpha]$.

Since $\Delta(f) = -23$ is square-free, it means that $\mathbb{Z}[\alpha]$ is integral.

10. Let $K(\sqrt{D})$ with D is square-free. Prove that $\mathcal{O}_K = \mathbb{Z}[\sqrt{D}]$ if $D \equiv 2, 3 \pmod{4}$ and $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{D}}{2}]$ if $D \equiv 1 \pmod{4}$.

Consider the discriminant of an order $\mathbb{Z}[\sqrt{D}]$ is given by

$$\mathfrak{d}(\mathbb{Z}[\sqrt{D}]) = 4D$$

The only value of $[\mathcal{O}_K : \mathbb{Z}[\sqrt{D}]]$ is 1 or 2. Suppose that $D \equiv 2, 3 \pmod{4}$ with $[\mathcal{O}_K : \mathbb{Z}[\sqrt{D}]] = 2$. Then we have an element

$$y = \frac{a + b\sqrt{D}}{2}$$

is integral. We have a minimal polynomial

$$f(x) = (x - y)(x - \sigma(y)) = x^2 - ax + \frac{a^2 - b^2 D}{4}$$

It must have $\frac{a^2 - b^2 D}{4}$ be an integer. It is impossible.

11. Find the Galois group of $\mathbb{Q}(\alpha, \sqrt{-23})/\mathbb{Q}$.

Clearly it permutes the roots of f . Since $f(x)$ is irreducible then it must be transitive. The transitive subgroup of S_3 are A_3 or S_3 . It can not be A_3 since $[\mathbb{Q}[\alpha, \sqrt{-23}] : \mathbb{Q}] = 6$. Also, the determinant $d = -23$ implies it admits two complex roots. consider the conjugate action, it is a odd permutation. Therefore, the galois group cannot be A_3 .

12. Analysis the integrality of $M = \mathbb{Z}[\alpha, \frac{1+\sqrt{-23}}{2}]$.

We consider the field extension $\mathbb{Q}(\alpha, \sqrt{-23})/\mathbb{Q}(\alpha)$ with minimal polynomial $g(x) = x^2 - x + 6$. We consider the determinant of g is given by

$$\Delta = -23$$

Therefore the determinant of M is given by calculate the determinant of $\{\alpha, \alpha^2, \alpha^3, \omega \cdot \alpha, \dots\}$ where we let $\omega = \frac{1+\sqrt{-23}}{2}$. Then we have

$$\Delta_M = (-23)^2 \cdot (-23)^3 = -23^5.$$

Let $L = \mathbb{Q}(\alpha, \sqrt{-23})$, $K_1 = \mathbb{Q}(\sqrt{-23})$, $K_2 = \mathbb{Q}(\alpha)$. Now we compute the determinant of field extension. I claim that the field extension $\mathbb{Q}(\alpha, \sqrt{-23})/\mathbb{Q}(\sqrt{-23})$ is unramified. Then $\mathfrak{D}(L/K_1) = \mathcal{O}_L$ and thus $\mathfrak{d}(L/K_1) = \mathcal{O}_{K_1}$. Consider the tower of Different

$$\mathfrak{D}(L/\mathbb{Q}) = \mathfrak{D}(L/K_1) \cdot \mathfrak{D}(K_1/\mathbb{Q})$$

and act on $N_{L/\mathbb{Q}}$ on both side. We then have

$$\mathfrak{d}(L/\mathbb{Q}) = N_{K_1/\mathbb{Q}}(\mathfrak{d}(L/K_1)) \cdot \mathfrak{d}(K_1/\mathbb{Q})^{[L:K_1]}.$$

And we have

$$\mathfrak{d}(L/\mathbb{Q}) = (-23)^3$$

Now we prove our claim.

First, you can check $K_1/\mathbb{Q}$ and $K_2/\mathbb{Q}$ are only ramified at 23. Therefore $L/\mathbb{Q}$ only ramified at 23.

Then you check $\mathbb{Q}_{23}(\alpha, \sqrt{-23})/\mathbb{Q}_{23}$ has degree 2. It implies that ramification index cannot larger than 2. Therefore L/K_1 is unramified.

13. Analysis the decomposition of (23) from the tower $\mathbb{Q} \rightarrow K_1 \rightarrow L$.

Consider $\mathbb{Q}_{23}(\sqrt{-23})/\mathbb{Q}_{23}$ with a minimal polynomial $g(x)$. The newton polygon of $g(x)$ tells its ramification index is 2. And since $K_1/\mathbb{Q}$ is quadratic extension, it must be totally ramified at (23) over K_1 . Now suppose that

$$23 \cdot K_1 = \mathfrak{p}^2.$$

We then consider the $K_1(\alpha)_{\mathfrak{p}}/(K_1)_{\mathfrak{p}}$ with a minimal polynomial $f(x)$. Consider the polynomial over residue field

$$x^3 - x + 1 \pmod{\mathfrak{p}}.$$

Since $f(3) \equiv 0 \pmod{\mathfrak{p}}$ and $f'(3) = 26 \not\equiv 0 \pmod{\mathfrak{p}}$. We obtain that $x = 3$ be the simple root of $f(x)$ according to Hensel's lemma. According to the symmetric of galois extension we then have

$$23 \cdot \mathcal{O}_L = (\mathfrak{P}_1 \mathfrak{P}_2 \mathfrak{P}_3)^2$$

14. Analysis the decomposition of (23) by using $L \otimes_{\mathbb{Q}} \mathbb{Q}_{23}$.

Basically, we use the fact that

$$L \otimes_{\mathbb{Q}} \mathbb{Q}_{23} = (K_1 \otimes_{\mathbb{Q}} \mathbb{Q}_{23}) \otimes_{\mathbb{Q}_{23}} (K_2 \otimes_{\mathbb{Q}} \mathbb{Q}_{23})$$

Analysis $K_1/\mathbb{Q}$ we now that $g(x)$ split over residue field. Then we obtain that

$$K_1 \otimes_{\mathbb{Q}} \mathbb{Q}_{23} = \mathbb{Q}_{23}(\sqrt{-23}).$$

Next we consider $K_2/\mathbb{Q}$ since

$$f(x) \equiv (x-3)(x-10)^2 \pmod{23}$$

Let $x = y + 10$, we find that

$$f(y) = x^3 + 30x^2 + 299x + 991$$

with Newton polygon

$$(0, 1) \rightarrow (0, 1) \rightarrow (0, 0) \rightarrow (0, 0)$$

It means that $K_2/\mathbb{Q}$ must ramified, then we obtain that

$$K_2 \otimes_{\mathbb{Q}} \mathbb{Q}_{23} \simeq \mathbb{Q}_{23} \times E_2$$

Where E_2 denotes an totally ramified extension of $\mathbb{Q}_{23}$ with $e = 2, f = 1$. According to the classification of local field extension, we claim it is equivalent to $\mathbb{Q}_{23}(\sqrt{-23})$. Therefore we then have

$$\begin{aligned} L \otimes_{\mathbb{Q}} \mathbb{Q}_{23} &= \mathbb{Q}_{23}(\sqrt{-23}) \otimes_{\mathbb{Q}_{23}} (\mathbb{Q}_{23} \times \mathbb{Q}_{23}(\sqrt{-23})) \\ &= \mathbb{Q}_{23}(\sqrt{-23}) \times \mathbb{Q}_{23}(\sqrt{-23}) \times \mathbb{Q}_{23}(\sqrt{-23}). \end{aligned}$$

15. Analysis the decomposition group and inertia group of (23).

We have

$$D_{\mathfrak{P}} = \text{Gal}(\mathbb{Q}_{23}(\sqrt{-23})/\mathbb{Q}_{23}) = \mathbb{Z}/2\mathbb{Z}$$

and exact sequence

$$0 \rightarrow I_{\mathfrak{P}} \rightarrow D_{\mathfrak{P}} \rightarrow \text{Gal}(\kappa(\mathfrak{P})/\mathbb{F}_{23}) \rightarrow 0$$

Since $\kappa(\mathfrak{P})/\mathbb{F}_{23}$ is trivial then we have $I_{\mathfrak{P}} = D_{\mathfrak{P}} = \mathbb{Z}/2\mathbb{Z}$.

16. Analysis the decomposition of (2).

We consider $L \otimes_{\mathbb{Q}} \mathbb{Q}_2$. Since $g(x)$ split over $\mathbb{F}_2$, it means that

$$K_1 \otimes_{\mathbb{Q}} \mathbb{Q}_2 = \mathbb{Q}_2 \times \mathbb{Q}_2.$$

Since $x^3 - x + 1$ has not root. It means that the inertia degree $f = 3$. Therefore we have

$$K_2 \otimes_{\mathbb{Q}} \mathbb{Q}_2 = \mathbb{Q}_{2^3}.$$

Then we have

$$L \otimes_{\mathbb{Q}} \mathbb{Q}_2 = \mathbb{Q}_{2^3} \times \mathbb{Q}_{2^3}.$$

17. Calculate the Frobenius element corresponding to (2).

We find it over extension $\mathbb{Q}_{2^3}/\mathbb{Q}_2$. Consider its residue field is $\mathbb{F}_{2^3}$. The Frobenius map $x \mapsto x^2$ has order 3 in this residue field. It means $\text{Frob}_{\mathfrak{P}}$ in $\text{Gal}(L/\mathbb{Q})$ must have order 3. The only case is (123).

18. Analysis the decomposition of (59) over L .

Similarly, we again Consider $\mathbb{Q}_{59}(\alpha, \sqrt{-23})/\mathbb{Q}_{59}$. Consider that

$$f(x) \equiv (x - 4)(x - 13)(x - 42) \pmod{59}$$

and we have

$$g(x) = (x - 3)(x + 3) \pmod{59}.$$

It means that $f(\mathbb{Q}_{59}(\alpha)/\mathbb{Q}_{59}) = f(\mathbb{Q}_{59}(\sqrt{-23})/\mathbb{Q}_{59}) = 1$. Consider that it is unramified, then we obtain it is totally split.

19. The extension $\mathbb{Q}_l(\zeta_p)/\mathbb{Q}_l$ is unramified for $l \neq p$.

Consider that $\zeta_p^p = 1$, then we have $v(\zeta_p^p) = 0$, which means that the value group is not change.

20. The extension $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ is totally ramified.

Since $x^p - 1 \equiv (x - 1)^p \pmod{p}$, it means that $f = 1$.

Hence we consider that

$$x^{p-1} + \binom{p}{1}x^{p-2} + \binom{p}{2}x^{p-3} + \cdots + p$$

It is irreducible over $\mathbb{Q}_p$ according to Dumas's criterion. It implies that the extension of local field is totally ramified.

21. Let $K = \mathbb{Q}(\sqrt[3]{m})$ with $m = ab^2$. Let $\alpha = \sqrt[3]{ab^2}$ and $\beta = \sqrt[3]{a^2b}$. Calculate the determinant of field K .

We use a fact that $K/\mathbb{Q}$ ramified over p iff. $p|d_K$. Consider $K = \mathbb{Q}(\alpha) = \mathbb{Q}(\beta)$ with minimal polynomials $f(x) = x^3 - ab^2$ and $g(x) = x^3 - a^2b$. Newton polygon tells every prime number p such that $p|ab$ is totally ramified. Therefore d_K must contains factors a, b . We consider the determinant of an order

$$d(1, \alpha, \beta) = -27a^2b^2$$

Since a, b are square free and the different between the discriminant of an order and field is square factor. Therefore, we claim there are two possibility

$$d_K = -27a^2b^2 \text{ or } -3a^2b^2.$$

When $3|a$ then 3 is totally ramified. One fact that we have is

$$3 \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]].$$

Suppose that $(3) = \mathfrak{p}^3$ and $v_{\mathfrak{p}}(\alpha) = k$. Then we have

$$3v_{\mathfrak{p}}(\alpha) = v_{\mathfrak{p}}(a) = 3.$$

Suppose that $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3g$, then we have

$$(3g)^{-1}\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K$$

which is impossible.

For same reason, if $3|b$, we also have

$$3 \nmid [\mathcal{O}_K : \mathbb{Z}[\beta]].$$

Consider that

$$[\mathcal{O}_K : \mathbb{Z}[\alpha, \beta]][[\mathcal{O}_K : \mathbb{Z}[\alpha]]]$$

or

$$[\mathcal{O}_K : \mathbb{Z}[\alpha, \beta]][[\mathcal{O}_K : \mathbb{Z}[\beta]]].$$

we have

$$3 \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha, \beta]].$$

Since $[\mathcal{O}_K : \mathbb{Z}[\alpha, \beta]]$ must equal to 1 or 3 and 3 is impossible. Then we find that if $3|ab$ then $d_K = -27a^2b^2$.

22. Prove that $\mathbb{Q}(\zeta_n + \zeta_n^{-1})$ is the maximal real subfield of $\mathbb{Q}(\zeta_n)$.

Consdier that ζ_n is the locus of

$$x^2 - (\zeta_n + \zeta_n^{-1})x + 1$$

23. The decomposition of cyclotomic polynomial

$$\Phi_n(x) = \prod_{d|n} (x^{n/d} - 1)^{\mu(d)}$$

an examples: $x^6 - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_6(x)$.

24. Give the relation between the number of complex embedding r_2 with determinant of f .

Let Δ be the determinant then we have

$$\Delta = \prod (\alpha_i - \alpha_j)^2$$

We have $(\alpha_i - \bar{\alpha}_i) < 0$, therefore we then have

$$\text{sgn}(\Delta) = (-1)^{r_2}$$

25. Calculate the class group of $\mathbb{Q}(\sqrt{-5})$.

Since $-5 \equiv 2 \pmod{4}$, we have $d_K = 4D$ and

$$\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}].$$

It has $r_1 = 0$ and $r_2 = 2$. We calculate the Minkowski bound

$$M_K = \frac{n!}{n^n} \left(\frac{4}{\pi}\right)^{r_2} \sqrt{|d_K|} \approx 2.84$$

Therefore, we just need to consider (2). According to Dedekind-Kummer theorem, the decomposition of

$$x^2 + 5 \equiv (x + 1)^2 \pmod{2}$$

implies that (2) is totally ramified

$$(2) = \mathfrak{p}^2$$

Where $\mathfrak{p} = (2, 1 + \sqrt{-5})$. Therefore $\text{Cl}(K)$ is generated by $[\mathfrak{p}]$. We just need to show that $\mathfrak{p}$ is not principle. Suppose that

$$(a + b\sqrt{-5}) = \mathfrak{p}$$

Act on norm on both side we have

$$2 = a^2 + 5b^2.$$

It has no integer solution. Therefore we have $\text{Cl}(K) = \mathbb{Z}/2\mathbb{Z}$.

26. Suppose that $N(\mathfrak{p}) = l$ where $\mathfrak{p}$ is a prime ideal and l is a prime number. Then if $N(\alpha) = l$, then we have

$$(\alpha) = \mathfrak{p}.$$

27. Consider the transitive subgroup of S_4 .

It only contains

28. Consider the Kummer extension of $\mathbb{Q}_p$

29. Find the fundamental

Class Field Theory

Idele class group

The Idele group is defined by

$$\mathbb{I}_L = \prod'_{v \in V} L_v^\times = \{(\alpha_v)_{v \in V} \mid \alpha_v \in K_v^\times, \text{all but finitely many } v, \alpha_v \in U_v\}$$

If we consider the field extension L/K , then we have

$$\mathbb{I}_L = \prod_{v \in V_K} \left(\prod_{w|v} L_w^\times \right)$$

Idele class group is defined by $\mathbb{I}_L/L^\times$.

There is a exact sequence

$$1 \rightarrow L^\times \rightarrow \mathbb{I}_L \rightarrow C_L \rightarrow 1.$$

invariant functor

Let G_K be the absolute galois group. Invariant functor $(\cdot)^{G_K}$ can be written by

$$(M)^{G_K} = \text{Hom}_{\mathbb{Z}[G_K]}(\mathbb{Z}, M).$$

for arbitrary G_K -module M .

Kummer exact sequence

Let K be a field such that $\text{char}(K) \nmid n$. We have Kummer exact sequence

$$1 \rightarrow \mu_n(K) \rightarrow \bar{K}^\times \xrightarrow{(\cdot)^n} \bar{K}^\times \rightarrow 1$$

and

$$1 \rightarrow \mu_n(K) \rightarrow \mathcal{O}_{\bar{K}}^\times \xrightarrow{(\cdot)^n} \mathcal{O}_{\bar{K}}^\times \rightarrow 1.$$

It induces long exact sequences given by

$$1 \rightarrow \mu_n(K) \rightarrow \bar{K}^\times \xrightarrow{(\cdot)^n} \bar{K}^\times \rightarrow H^1(G_K, \mu_n(K)) \rightarrow H^1(G_K, \bar{K}^\times) \rightarrow \dots$$

and

$$1 \rightarrow \mu_n(K) \rightarrow \mathcal{O}_{\bar{K}}^\times \xrightarrow{(\cdot)^n} \mathcal{O}_{\bar{K}}^\times \rightarrow H^1(G_K, \mu_n(K)) \rightarrow H^1(G_K, \mathcal{O}_{\bar{K}}^\times) \rightarrow \dots$$

Hilbert's 90 theorem

We have Hilbert's 90 theorem.

1. $H^1(G_K, \bar{K}^\times) = 0$.
2. L/K be the finite galois extension over a local field K . Then we have $H^1(G, L^\times) = 0$.
3. Additivity from above. we have $H^i(G, L) = 0$ for all $i \geq 1$.
4. If L/K is the finite galois extension over global field. we have $H^1(G, L^\times) = 0$.
5. We have $H^1(G, \mathbb{I}_L) = 0$ and $H^1(G, C_L) = 0$.

Shapiro's Lemma

Shapiro's lemma claim that

$$H^i(G, \text{Ind}_H^G(M)) \cong H^i(H, M) \quad (\forall i \geq 0)$$

therefore we have

$$H^i(G, \mathbb{I}_L) \cong \bigoplus_v H^i(D_{\mathfrak{p}}, L_w^\times) \quad (\forall i \geq 0)$$

The dimension of cohomology

Consider K is a local nonarchimedean field with finite G_K -module M . Then

$$H^i(G_K, M) = 0$$

for $i \geq 3$.

The meaning of cohomology

The zero order cohomology

$$H^0(G_K, M) = M^{G_K}$$

means the element of M which is contained over base field.

Tate Duality

We consider local field K .

$$\mu_n = \{\zeta \in \bar{K}^\times \mid \zeta^n = 1\}$$

Let μ_∞ be the inductive limit of μ_n .

Let M be a finite G_K -module. Define Cartier duality by

$$M^\vee = \underline{\text{Hom}}_{\mathbb{Z}}(M, \mu_\infty)$$

For $f \in M^\vee$, we have $\sigma \cdot f = \sigma f \sigma^{-1}$. As a set it is equal to $\text{Hom}_{\mathbb{Z}}(M, \mu_\infty)$.

Hilbert Symbol and Artin Reciprocity Law

The Hilbert Symbol and artin Reciprocity law are two essential and basic concept of Class Field Theory.

Consider local field K with $\mu_n \subseteq K$. The Hilbert symbol is defined by

$$(\cdot, \cdot)_K : (K^\times / K^{\times n}) \times (K^\times / K^{\times n}) \rightarrow \mu_n$$

$$(a, b)_K \mapsto \frac{\sigma_b(\sqrt[n]{a})}{\sqrt[n]{a}}$$

where

$$\sigma_b : K^\times \rightarrow G_K^{\text{ab}}$$

is the Artin map.

The meaning of Hilbert symbol is quite simple. That is

$$(a, b)_K = 1 \iff a \in N_{L/K}(L^\times) \iff \text{The equation } x^n - a = 0 \text{ has solution over } L$$

where $L = K(\sqrt[n]{b})$

local/global class field, tate duality, Hilbert class, reciprocity, gal cohomology, kummer exact sequence Herbrand

The formula

30. Prove that $x^2 + 5 = y^3$ does not have an integer solution.

31. Consider the quadratic Hilbert symbol, prove that $(-1, -1)_2 = -1$ and $(2, 5)_2 = -1$. Suppose that $(2, 5) = 1$. It means the norm function

$$a^2 - 2b^2 = 5$$

has a solution over $\mathbb{Q}_2$. First $a, b \in \mathbb{Z}_2$, if not we obtain that $v_2(5) < 0$. Therefore we can consider mod 8. The only quadratic element mod 8 are 1 or 4. The only possible mod 8 of LHS are 0, 1, 2, 4, 6, 7. The same as $(-1, -1)_2$.

32. Consider the decomposition of $\mathbb{Q}_2^\times$ and $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$. Deduce the Hilbert symbol for $\mathbb{Q}_2$. We have

$$\mathbb{Q}_2^\times \simeq \mathbb{Z} \times \mathbb{Z}_2^\times \simeq \mathbb{Z} \times (1 + 4\mathbb{Z}_2) \times \mu_2.$$

Therefore we have

$$\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2} = \langle -1, 3, 5 \rangle \simeq (\mathbb{Z}/2\mathbb{Z})^3.$$

33. Let $L = \mathbf{Q}(\sqrt{2})$. Determine whether the norm equation $N_{L/\mathbf{Q}}(z) = 3$ has a solution $z \in L^\times$. Equivalently, decide whether $x^2 - 2y^2 = 3$ has a solution in $\mathbf{Q}$. Use local Hilbert symbols.

Since $L/\mathbf{Q}$ is cyclic, the Hasse norm theorem says

$$3 \in N_{L/\mathbf{Q}}(L^\times) \iff 3 \in N_{L_v/\mathbf{Q}_v}(L_v^\times) \text{ for all places } v.$$

For the quadratic extension $L = \mathbf{Q}(\sqrt{2})$, the local norm condition is

$$3 \in N_{\mathbf{Q}_v(\sqrt{2})/\mathbf{Q}_v}(\mathbf{Q}_v(\sqrt{2})^\times) \iff (2, 3)_v = 1.$$

We compute $(2, 3)_v$. For odd p , write

$$a = p^\alpha u, \quad b = p^\beta v,$$

with $u, v \in \mathbf{Z}_p^\times$. Then

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right)^\beta \left(\frac{\bar{v}}{p}\right)^\alpha.$$

If $p \neq 2, 3$, then $2, 3 \in \mathbf{Z}_p^\times$, so

$$(2, 3)_p = 1.$$

For $p = 3$,

$$2 = 3^0 \cdot 2, \quad 3 = 3^1 \cdot 1,$$

hence

$$(2, 3)_3 = \left(\frac{2}{3}\right) = -1.$$

For $p = 2$, use the formula

$$(a, b)_2 = (-1)^{\epsilon(u)\epsilon(v) + \alpha\omega(v) + \beta\omega(u)},$$

where

$$\epsilon(u) = \frac{u-1}{2} \pmod{2}, \quad \omega(u) = \frac{u^2-1}{8} \pmod{2}.$$

Since

$$2 = 2^1 \cdot 1, \quad 3 = 2^0 \cdot 3,$$

we get

$$(2, 3)_2 = (-1)^{\omega(3)} = (-1)^{(9-1)/8} = -1.$$

At the real place,

$$(2, 3)_\infty = 1$$

because $2 > 0$ and $3 > 0$.

Therefore

$$(2, 3)_2 = (2, 3)_3 = -1, \quad (2, 3)_v = 1 \quad \text{for } v \neq 2, 3.$$

In particular,

$$(2, 3)_2 \neq 1,$$

so 3 is not a local norm from $\mathbf{Q}_2(\sqrt{2})$. Hence it is not a global norm.

Thus

$$N_{\mathbf{Q}(\sqrt{2})/\mathbf{Q}}(z) = 3 \text{ has no solution in } \mathbf{Q}(\sqrt{2})^\times.$$

Equivalently,

$$x^2 - 2y^2 = 3 \text{ has no solution in } \mathbf{Q}.$$

Finally, Hilbert reciprocity is still satisfied:

$$\prod_v (2, 3)_v = (2, 3)_2 (2, 3)_3 = (-1)(-1) = 1.$$

10. Let $K = \mathbf{Q}_3$, and let

$$L = K(\sqrt{2}).$$

Let

$$\chi \in H^1(G_K, \mathbf{Q}/\mathbf{Z}) = \text{Hom}_{\text{cont}}(G_K, \mathbf{Q}/\mathbf{Z})$$

be the quadratic character corresponding to L/K , normalized so that the nontrivial element has value $1/2 \in \mathbf{Q}/\mathbf{Z}$.

Using the cohomological Artin reciprocity pairing

$$H^1(G_K, \mathbf{Q}/\mathbf{Z}) \times K^\times \longrightarrow \mathbf{Q}/\mathbf{Z},$$

$$(\chi, b) \longmapsto \text{inv}_K(\delta(\chi) \cup b),$$

where

$$\delta(\chi) \in H^2(G_K, \mathbf{Z})$$

comes from

$$0 \rightarrow \mathbf{Z} \rightarrow \mathbf{Q} \rightarrow \mathbf{Q}/\mathbf{Z} \rightarrow 0,$$

and

$$\delta(\chi) \cup b \in H^2(G_K, \bar{K}^\times) = \text{Br}(K),$$

compute

$$\text{inv}_K(\delta(\chi) \cup 3).$$

Then compare your answer with

$$\chi(\text{Art}_K(3)).$$

Solution.

Since 2 is not a square modulo 3, the extension

$$L = \mathbf{Q}_3(\sqrt{2})$$

is the unramified quadratic extension of $\mathbf{Q}_3$.

By local Artin reciprocity, if we choose the convention

$$\text{Art}_K(3) = \text{Frob}_3,$$

then $\text{Art}_K(3)$ is the nontrivial element of

$$\text{Gal}(L/K) \cong \mathbf{Z}/2\mathbf{Z}.$$

Therefore

$$\chi(\text{Art}_K(3)) = \frac{1}{2}.$$

Now compute the cohomological side. The class

$$\delta(\chi) \cup 3 \in H^2(G_K, \bar{K}^\times) = \text{Br}(K)$$

is the Brauer class corresponding to the quadratic Hilbert symbol

$$(2, 3)_3.$$

For odd p , if

$$a = p^\alpha u, \quad b = p^\beta v,$$

with $u, v \in \mathbf{Z}_p^\times$, then

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right)^\beta \left(\frac{\bar{v}}{p}\right)^\alpha.$$

Here

$$2 = 3^0 \cdot 2, \quad 3 = 3^1 \cdot 1.$$

Hence

$$(2, 3)_3 = \left(\frac{2}{3}\right) = -1.$$

The value $-1 \in \mu_2$ corresponds to

$$\frac{1}{2} \in \frac{1}{2}\mathbf{Z}/\mathbf{Z} \subset \mathbf{Q}/\mathbf{Z}.$$

Therefore

$$\text{inv}_K(\delta(\chi) \cup 3) = \frac{1}{2}.$$

Thus

$$\boxed{\text{inv}_K(\delta(\chi) \cup 3) = \chi(\text{Art}_K(3)) = \frac{1}{2}.}$$

This verifies the cohomological form of local Artin reciprocity in this example:

$$\boxed{\text{inv}_K(\delta(\chi) \cup b) = \chi(\text{Art}_K(b)).}$$

Problem.

Let

$$L = \mathbf{Q}(\zeta_7)^+$$

be the maximal real subfield of $\mathbf{Q}(\zeta_7)$. Then

$$[L : \mathbf{Q}] = 3$$

and

$$\text{Gal}(L/\mathbf{Q}) \cong \mathbf{Z}/3\mathbf{Z}.$$

This extension is not a Kummer extension over $\mathbf{Q}$, since

$$\mu_3 \not\subset \mathbf{Q}.$$

(1)

Use global class field theory to compute

$$C_{\mathbf{Q}}/N_{L/\mathbf{Q}}C_L.$$

(2)

Interpret this quotient using Tate cohomology.

(3)

Determine the Artin symbols of the primes 2 and 13, and decide whether

$$2, 13 \in N_{L/\mathbf{Q}}(L^\times).$$

Solution.

Since

$$L = \mathbf{Q}(\zeta_7)^+$$

is the real subfield of the cyclotomic field $\mathbf{Q}(\zeta_7)$, we have

$$\text{Gal}(L/\mathbf{Q}) \cong (\mathbf{Z}/7\mathbf{Z})^\times / \{\pm 1\}.$$

But

$$(\mathbf{Z}/7\mathbf{Z})^\times = \{1, 2, 3, 4, 5, 6\} \cong \mathbf{Z}/6\mathbf{Z},$$

so

$$(\mathbf{Z}/7\mathbf{Z})^\times / \{\pm 1\} \cong \mathbf{Z}/3\mathbf{Z}.$$

By global class field theory, the Artin reciprocity map gives an isomorphism

$$C_{\mathbf{Q}}/N_{L/\mathbf{Q}}C_L \xrightarrow{\sim} \text{Gal}(L/\mathbf{Q}).$$

Therefore

$$\boxed{C_{\mathbf{Q}}/N_{L/\mathbf{Q}}C_L \cong \mathbf{Z}/3\mathbf{Z}.$$

Now let

$$G = \text{Gal}(L/\mathbf{Q}).$$

Tate cohomology gives

$$\hat{H}^0(G, C_L) = C_{\mathbf{Q}}/N_{L/\mathbf{Q}}C_L.$$

The fundamental class

$$u_{L/\mathbf{Q}} \in H^2(G, C_L)$$

induces, by Tate–Nakayama duality, an isomorphism

$$\hat{H}^{-2}(G, \mathbf{Z}) \xrightarrow{\cup u_{L/\mathbf{Q}}} \hat{H}^0(G, C_L).$$

But

$$\hat{H}^{-2}(G, \mathbf{Z}) \cong H_1(G, \mathbf{Z}) \cong G^{\text{ab}}.$$

Since G is already abelian,

$$G^{\text{ab}} = G.$$

Hence

$$\hat{H}^0(G, C_L) \cong G.$$

This recovers the class field theory isomorphism

$$C_{\mathbf{Q}}/N_{L/\mathbf{Q}}C_L \cong \text{Gal}(L/\mathbf{Q}).$$

Now compute Frobenius elements. For a prime $\ell \neq 7$, the Artin symbol is given by

$$\text{Frob}_{\ell} = [\ell] \in (\mathbf{Z}/7\mathbf{Z})^{\times}/\{\pm 1\}.$$

For $\ell = 2$,

$$2 \not\equiv \pm 1 \pmod{7}.$$

The class of 2 has order 3 in

$$(\mathbf{Z}/7\mathbf{Z})^{\times}/\{\pm 1\}.$$

Thus

$$\text{Frob}_2$$

is a generator of

$$\text{Gal}(L/\mathbf{Q}) \cong \mathbf{Z}/3\mathbf{Z}.$$

Hence 2 does not split completely in L .

For $\ell = 13$,

$$13 \equiv -1 \pmod{7}.$$

Thus the class of 13 is trivial in

$$(\mathbf{Z}/7\mathbf{Z})^{\times}/\{\pm 1\}.$$

Therefore

$$\text{Frob}_{13} = 1.$$

Hence 13 splits completely in L .

Finally, since $L/\mathbf{Q}$ is cyclic, the Hasse norm theorem applies:

$$a \in N_{L/\mathbf{Q}}(L^{\times}) \iff a \in N_{L_v/\mathbf{Q}_v}(L_v^{\times}) \quad \text{for all } v.$$

For a rational prime $\ell \neq 7$, the local norm condition at ℓ forces the valuation $v_{\ell}(\ell) = 1$ to be divisible by the residue degree f_{ℓ} . Thus ℓ can be a local norm at ℓ only if

$$f_{\ell} = 1,$$

i.e. only if ℓ splits completely in L .

Since 2 does not split completely,

$$2 \notin N_{L/\mathbf{Q}}(L^\times).$$

Since 13 splits completely and is locally a norm at every place, the Hasse norm theorem gives

$$13 \in N_{L/\mathbf{Q}}(L^\times).$$

Thus this example uses global class field theory and Tate cohomology, but not Kummer theory.